

making use of Big Data analytics in business, as compared to 17 per cent in 2015, which marks a tremendous growth in adoption.

Many companies are interested in data science to improve decision making, but businesses are mostly unaware of the need for planning and carrying out such activities. To make efficient use of data science, the primary requirement is skilled personnel, i.e., data scientists. Such professionals are responsible for collecting, analysing and interpreting large amounts of data to identify ways to help the business improve operations and gain a competitive edge over rivals. The best resources and infrastructure are also required to carry out all data analysis activities in a smooth manner. In addition, it is necessary to identify possible sources of data and permissions as well as the methods needed to acquire the data. The next step is learning and building data science skills. Statistical, modelling, programming, visualisation, machine learning and data mining skills are required to carry out data science activities. The final step is to take the relevant proactive action based on the insights from data analysis.

The major hurdle in data science is the availability of data. Data collection as well as data structuring is essential, before data can be made useful. Then the data has to be cleaned, analysed and processed into a suitable model with efficient presentation.

The following reasons highlight the importance of data science and Big Data analytics for today's enterprises.

- **Effective decision making:** Data science and Big Data analytics act as trusted advisors to the organisation, particularly for effective strategic planning. They provide a strong base for the management to enhance its analytical abilities and the overall decision making process. They enable the measuring, recording and performance tracking metrics that allow the top management to set

new goals and objectives for profit maximisation.

- **Identification of competitive trends:** Data analytics enables organisations to determine the patterns within large data sets. This is very useful in identifying new and upcoming market trends. Once the trends are identified, these can become a useful parameter for the enterprise to gain a competitive advantage by introducing new products and services in the market to gain an edge over rivals.
- **Efficiency in handling core tasks and issues:** By making the employees aware of the benefits of the organisation's analytics product, data science can enhance job efficiency. Working closely with company goals, the employees will be able to direct more resources towards core tasks and issues at every stage, which will enhance the operational efficiency of the business.
- **Promoting low risk data-driven action plans:** Big Data analytics makes it possible for every SME to take action based on quantifiable, data-driven evidence. With such actions, businesses can reduce unnecessary tasks and curtail certain risks to a remarkable level.
- **Selecting a target audience:** Big Data analytics plays a key role in providing more insight into customer preferences and expectations. With deeper analysis, companies can identify target audiences and can propose new customer-driven and customer-oriented products and services.

The growing adoption of data science and analytics in enterprises

In the past three years, Big Data analytics has seen a significant increase in usage. The following are some interesting statistics:

- Reporting, dashboards, advanced visualisation, end-user 'self-service' and data warehousing are the top

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